LogScope

Turning Application Logs into Actionable Insights

A Lightweight Log Analysis and Error Diagnosis Platform

Advanced Python Programming Project by P. Ganesh Prakadesh



THE PROBLEM

Finding errors shouldn’t take hours

Applications generate large volumes of logs, while developers often search them manually. Several visible errors may point back to one underlying issue.

RAW LOG STREAM

Timeout... retry... stack trace... failed request...

Many lines. Few relationships.

ORGANIZED VIEW

1 underlying pattern

Grouped errors, severity, context, and next steps.

Splunk and the ELK Stack are established options, yet setup, infrastructure, cost, or complexity may not fit every individual or small team. LogScope focuses on straightforward uploads, organized analysis, and diagnostic suggestions.

THE CORE WORKFLOW

From raw logs to useful insights

01

Upload

Upload and parse the log.

02

Group

Detect, classify, and group repeated error patterns.

03

Correlate

Relate events to rules and stored knowledge.

04

Explain

Display errors, statistics, explanations, and suggested fixes.

A database timeout followed by failed checkout requests may be related. Correlation suggests a relationship; it does not prove root cause.

Temporary log data is released under the privacy design after analysis.

ONE INTERFACE · MULTIPLE CATEGORIES

One interface, multiple log categories

LogScope brings parsing, severity analysis, grouping, correlation, rule-based suggestions, dashboard summaries, and reusable diagnostic knowledge into one view.



SQL / database

Connection failures, timeouts, query errors.



Minecraft

Crashes, plugins, server events.



Python

Exceptions and application failures.



Java

Exceptions and runtime errors.

Examples illustrate intended categories, not universal format coverage or guaranteed diagnosis.

AUDIENCE

Who benefits from LogScope?



Developers

Investigate Java or Python exceptions.



Server administrators

Investigate Minecraft crashes and plugin errors.



Database users

Investigate SQL errors and database failures.



Other users

IT support teams, students, and small software teams.

Intended benefits: less manual searching, organized troubleshooting, easier access to diagnostic information, and possible reuse of verified solutions.

No customer or cost-savings claims are made.

SYSTEM ARCHITECTURE

How the application is built

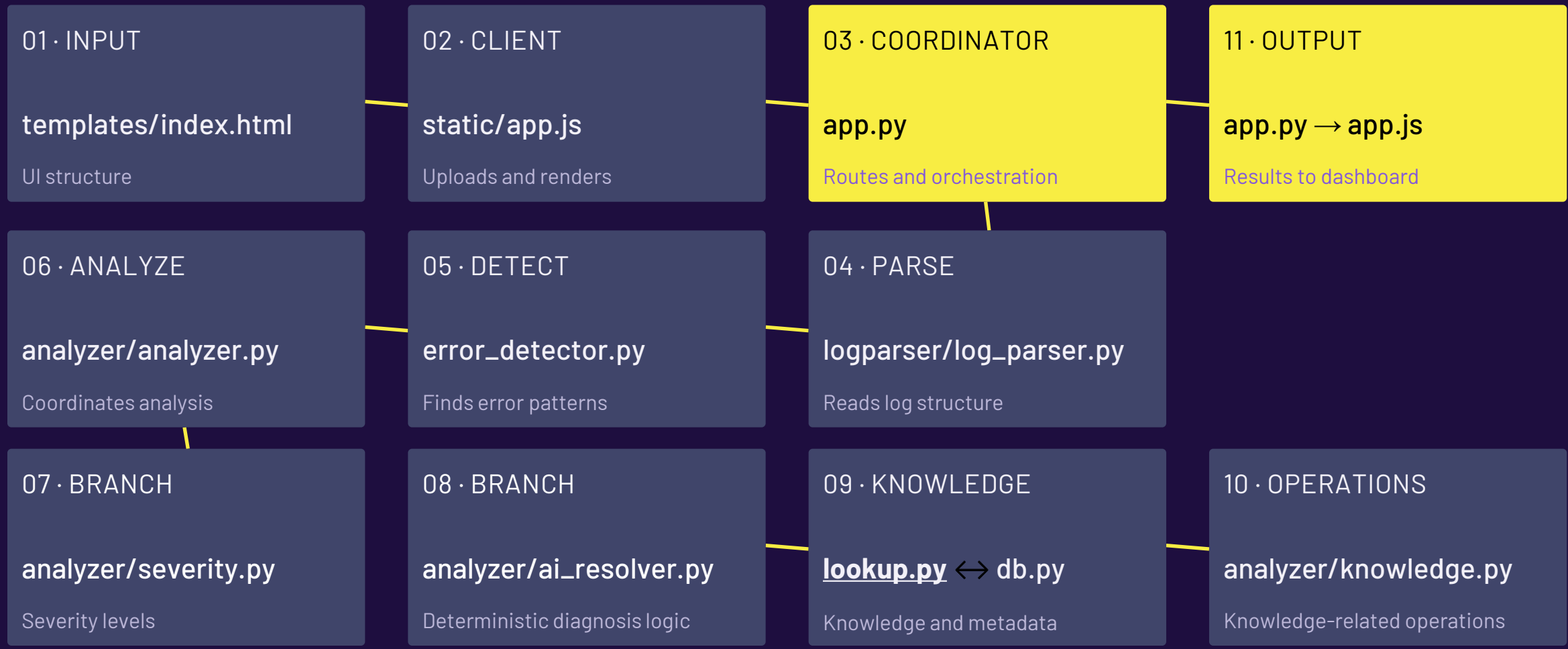


Uploaded logs are temporary analysis inputs. Persistent storage is for generalized, sanitized knowledge and metadata.

FILE ARCHITECTURE

Inside LogScope: how the Python files work together

Dependency view · concise relationships only · ai_resolver.py is deterministic diagnosis logic



PRIVACY-BY-DESIGN DIRECTION

Reusable knowledge without retaining raw logs

01 **Analyze**

Identify a recognizable pattern in the uploaded log.

02 **Retrieve**

Present relevant knowledge, a possible diagnosis, and a suggested fix.

03 **Review**

Allow confirmation in the intended verification workflow.

04 **Sanitize**

Sanitize reusable information before persistent storage; retrieve it in future analyses.

Temporary

Uploaded logs are processed for the current analysis.

Persistent

PostgreSQL knowledge stores generalized, sanitized knowledge and metadata—not raw logs.

Confirmation, database-write, and future-retrieval steps remain planned or subject to end-to-end verification unless tested. Protect credentials and sensitive data. Suggested causes are not guaranteed root causes.

THESIS POSITION

Current value and future direction

POTENTIAL BENEFITS

Reduce friction

- Less manual investigation
- Better organization
- Faster initial troubleshooting
- Reuse of verified solutions
- Lightweight modular architecture

LIMITATIONS

Stay precise

- Accuracy depends on log quality and available rules
- Unknown errors may lack reliable explanations
- Correlation does not establish causality
- Business benefits still need measurement

FUTURE IMPROVEMENTS

Build evidence

- Expand log support and correlation
- Strengthen verification and sanitization
- Add regression tests and large-file handling
- Measure accuracy and investigation time
- Improve security and retention controls

CLOSING THESIS

Making Log Troubleshooting Simpler

01 **Problem:** Large log files make error investigation time-consuming.

02 **Solution:** LogScope organizes logs, detects and groups errors, and suggests possible causes through rule-based analysis.

03 **Business value:** The application aims to reduce investigation effort and improve access to reusable diagnostic knowledge.



> “LogScope aims to turn complex application logs into clearer diagnostic information, helping users spend less time searching for errors and more time solving them.”

Thank You · Questions